

Raied Ahmed Nishat

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EDUCATION

BSc (Engg.) in Civil and Environmental Engineering

Jan 2020 - Sep 2025

Shahjalal University of Science and Technology (SUST), Sylhet 3114, Bangladesh
GPA: 3.64/4.00, Class Rank 5/55

Undergraduate Thesis: A Transfer Learning Approach for Soil Strength Prediction

Advisor: [Dr. Shriful Islam](#)

PROFESSIONAL EXPERIENCE

Lecturer (Adjunct)

Jan 2026 - Present

Department of Civil Engineering, Leading University, Sylhet-3112, Bangladesh

- **CE 3181: Structure III - Mechanics of Solids** Spring 2026
Course Description: Analysis of flexural and shear stresses, principal stresses, beam deflection (direct integration and area-moment methods), intermediate beam behavior, and column buckling for structural stability assessment.
- **CE 3132: Open Channel Flow Sessional** Spring 2026
Course Description: Laboratory investigation of open channel hydraulics, including flow measurement structures (weirs, sluice gates, Venturi, Parshall and cutthroat flumes), hydraulic jump, velocity distribution, specific energy/force, and estimation of Manning's roughness coefficient.
- **CE 4222: Environmental Engineering Sessional II** Spring 2026
Course Description: Applied hydraulic design of water supply, sewerage, and stormwater systems, including design of treatment units (bar screens, grit chambers, sedimentation tanks, trickling filters) with EPANET-based modeling and field evaluation.
- **CE 4116: Concrete Structures Sessional II** Summer 2026
Course Description: Design of a low-rise reinforced concrete building (wall system), Design of a multi-storied reinforced concrete building (beam-column system) including earthquake resistant design.
- **CE 4251: Traffic Planning and Management** Summer 2026
Course Description: Transportation planning process; management concepts; traffic accident investigations; city road and street networks; grade separation and interchanges, pedestrian and bicycle facilities. elements of traffic flow.
- **CE 3263: Building Services II: Plumbing** Summer 2026
Course Description: Principles and theories of, plumbing, water requirement, fire safety criteria, water requirement for fire safety, water sources, supply and distribution of water supply network, sewage and sewer system, elements of building plumbing, elements of sewerage systems, sanitation..

Junior Project Engineer

Dec 2025 - Jan 2026

Center for Research Testing and Consultancy (CRTC), SUST, Sylhet-3114

- Supervised on-site SPT operations at Sylhet Gas Field sites and conducted laboratory soil classification and strength testing to ensure reliable geotechnical data.
- Contributed to technical analysis and preparation of engineering reports for consultancy projects.

HONORS AND ACHIEVEMENTS

- Semifinalist, Hult Prize at SUST, 2023
- Chandpur District Parishad Higher Secondary Merit Scholarship
- Merit Scholarship, Secondary School Certificate
- Chandpur District Parishad Secondary School Merit Scholarship
- Ashutosh Education Scholarship, Brahmanbaria

RESEARCH EXPERIENCE

Research Assistant

Apr 2024 - Present

Department of Civil and Environmental Engineering, SUST

Advisor: Prof. Dr. Shriful Islam

- Laboratory-based experimental and computational studies on material and soil characterization.
- Applying machine learning in geotechnical modeling of soil strength and relevant parameters.
- Conducting characterization tests, data analysis, report and manuscript preparation for ongoing projects.

Collaborative Research Projects (journal manuscript in-preparation)

1. Performance Evaluation of Water Distribution Systems

Co-Authors: [Dr. Ram Krishna Mazumder](#) (Georgia Southern University)

Topic: This study evaluates various performance metrics, including topographic, hydraulic, water security and economic metrics for water networks.

2. Community-Level Scenario Seismic Risk and Resilience Analysis of Chittagong

Co- Authors: [Abul Hasnat](#) (Texas A&M University) and [Dr. Ram Krishna Mazumder](#) (Georgia Southern University)

Topic: Using scenario seismic hazards, this study evaluates seismic risk of buildings and post-earthquake recovery for Chittagong city. Utilizing the HAZUS framework, this study incorporates uncertainty considerations in damage, retrofitting, and recovery simulations.

3. Machine Learning in Soil Strength Prediction: A Systematic Review

Co-Authors: Fahmida Rahman (SUST), and Dr. Shfirul Islam (SUST)

Topic: First targeted systematic review of machine learning application in soil strength parameter prediction. Following PRISMA guided methodology for systematic reviews, this study compiles a comprehensive review of current literature trends, gaps, providing future directions for this field.

4. Uncertainty Aware Bayesian Physics Informed Neural Network for Probabilistic Slope Stability Classification

Co-Authors: Nabid Hasnat (Leading University, Sylhet)

Topics: Incorporating monotonic physics constraints in Bayesian Neural Networks for developing a physically consistent, uncertainty-aware machine learning framework for slope stability classification, providing a trustworthy, probabilistic risk-assessment tool that quantifies predictive uncertainty while strictly following soil mechanics.

5. Machine Learning-Based Prediction of Static Pile Load Capacity and Development of Empirical Equations for BNBC 2020 Refinements

Co-Authors: Sakib Bhuiyan (Leading University, Sylhet)

Topics: Developing an interpretable machine learning based framework for pile load capacity prediction and recommendation to BNBC 2020 for empirical equation refinement.

PUBLICATIONS

Journal Articles

1. **Nishat, R. A.**, Rahman, F., Hoque, N. M. R., & Islam, S. (2026). Transfer learning for undrained shear strength prediction: A case study with instance-based transfer. *Geosystems and Geoenvironment (Elsevier)*, 100519. <https://doi.org/10.1016/j.geogeo.2026.100519>
2. Islam S., **Nishat R. A.**, Sutradhar A., Hoque N. M. R., Rafi M. N. I. and Rana S. (202X). Effect of lime and alkali-activated LRF slag on geotechnical properties and microstructural behavior of kaolin clay, *International Journal of Geosynthetics and Ground Engineering (Springer)*. **(Under Review)**
3. Islam S., **Nishat R. A.**, Naba M.F., Chowdhury A.S., and Rafi M. N. I. (202X). Strength and mechanical behavior of fly ash and lime stabilized embankment soil reinforced with polyester fiber, *Indian Geotechnical Journal (Springer)*. **(Under Review)**
4. Joy M.R. and **Nishat R. A.** Hydroclimatic Trend Analysis and Machine Learning-Based Water Level Prediction for the Ganges (Padma) River Basin. *Acta Geophysica (Springer)*. **(Under Review)**
5. **Nishat R. A.** and Joy M.R. (2025). Investigating the Trends of Rainfall and Discharge along the Brahmaputra-Jamuna River System. *Technical Journal of River Research Institute*, 17(1). [\[ResearchGate\]](#)

Conference Papers

1. **Nishat R. A.**, Tusar A. N., Islam S., and Monni M. A. (2026). A Probabilistic Transformer-Based Machine Learning Model for Shallow Foundation Settlement Prediction. *8th International Conference on Civil Engineering for Sustainable Development 2026 (KUET), Khulna, February 5-7, 2026, 2026*. [\[ResearchGate\]](#)

TECHNICAL SKILLS

- **Machine Learning Methods:** Neural Networks, Transfer learning, Gradient Boosting, Random Forest, Bayesian/Probabilistic Modeling
- **Geotechnical & Scientific Software:** EPANET, SWMM, AutoCAD, ETABS, ArcGIS, QGIS
- **Programming & Productivity:** Python, TensorFlow/Keras, Jupyter Notebook, LaTeX, C (basic), MATLAB (basic)

LEADERSHIP AND SERVICE ACTIVITY

General Secretary | Aaj Muktomancho, SUST

2024 – 2025

- Led the 15th Executive Committee and served as convener of a 1,000+ attendee cultural concert, overseeing programs, partnerships, and public engagement.

Organizing Secretary | Association of Civil and Environmental Engineering, SUST

2024 – 2025

- Coordinated large-scale departmental events and facilitated collaboration among faculty, students, and industry stakeholders.

CERTIFICATES AND TRAINING

- **A complete guide to the key components of a research article.** Nov, 2025
Wiley Researcher Academy, Credential ID: cert_2nh3qw14
- **Introduction to GIS Mapping | Coursera.** Dec, 2024
University of Toronto, Credential ID: ZLMCG05UAA19

REFERENCES

Dr. Shriful Islam

Undergraduate Thesis Supervisor

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Shahjalal University of Science and Technology
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Dr. Ram Krishna Mazumder

Research Mentor and Collaborator

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